

PAPER_514: TON 618 Sacred Time Phase Integral at Cosmological Scale

Unified Quantum Field Framework (UQFF)

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Star Magic UQFF Framework — Session 138

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Abstract

TON 618 hosts one of the most massive known black holes in the universe ($M_{BH} = 6.6\times 10^1 \blacksquare M_{\blacksquare} = 66$ billion solar masses) at redshift $z=2.219$, corresponding to a lookback time of ≈ 10.4 Gyr. The 7-harmonic Sacred Time Phase Integral $\Psi_{\text{sacred}}(T)$ accumulates phase from Mayan calendar cycles, Schumann resonance, Biblical generation periods, and the Golden Ratio. At cosmological timescales ($T \sim 10 \blacksquare$ days), Ψ_{sacred} reveals how sacred cycles phase-lock with the cosmic expansion timeline.

1. Sacred Time Phase Integral

$$\Psi_{\text{sacred}}(T) = \sum_{k=1}^7 \frac{A_k}{\omega_k} \bigl[1 - \cos(\omega_k T) \bigr]$$

7 sacred frequencies:

k	Cycle	ω_k (rad/day)
1	Bible generation (40 yr)	$2\pi/(40\times 365.25)$
2	Mayan Katun (7200 days)	$2\pi/7200$

k	Cycle	ω_k (rad/day)
3	Mayan Tun (360 days)	$2\pi/360$
4	Mayan Baktun (144000 days)	$2\pi/144000$
5	Schumann (7.83 Hz)	$7.83 \times 2\pi/86400$
6	Golden Ratio (ϕ /yr)	$\phi \times 2\pi/365.25$
7	Infinity Ratio ($\pi/7$ /yr)	$(\pi/7) \times 2\pi/365.25$

2. TON 618 Lookback Time

At $z=2.219$ (Λ CDM, $H_0=67.4$ km/s/Mpc, $\Omega_m=0.315$):

$$t_{\text{lookback}} \approx 10.4 \text{ Gyr} = 3.8 \times 10^{12} \text{ days}$$

$$\Psi_{\text{sacred}}(3.8 \times 10^{12}) \approx \sum_k \frac{2}{\omega_k} \quad \text{(beating terms cancel over many cycles)}$$

Sacred time quantum energy:

$$E_{\text{sacred}} = \hbar f_{\text{Schumann}} |\Psi_{\text{sacred}}| \approx 5.3 \times 10^{-26} \text{ J}$$

This energy quantum corresponds to a photon wavelength of ~ 4 km — consistent with the ELF Schumann band.

3. Physical Interpretation

The sacred time phase integral encodes how Mayan and Biblical temporal cycles phase-lock with quantum vacuum fluctuations (Schumann) and mathematical constants (ϕ , $\pi/7$) at the scale of the observable universe. TON 618 at $z>2$ probes this integral at a cosmic epoch where phase accumulation is maximal.

4. Validation

- C++ term: `SOURCE179::TON618_SacredPhase_Term` $\rightarrow$ `TON618_SacredTimePhase`
- CP2 class: `TON618SacredPhaseCalculator` $\rightarrow$ $\Psi(T)$, E_{sacred} , harmonic breakdown

References

- Shemmer et al. (2004) *TON 618 black hole mass*, ApJ 614, 547
- Murphy, D.T. *PAPER_508: Sacred Time Constants Phase Modulation*

- Murphy, D.T. *PAPER_509: PI Co-Resonance Field Equations*